#6257/2017/ADCP
11/08/2017

#8431/2017/ADSP
03/08/2017

Indian Institute of Technology, Delhi

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CENTRAL RESEARCH FACILITY

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Ref.: IITD/TT/AKA/CFC03
1 August 2017

Re.: Minutes of CFC for approval

I am enclosing the minutes of the business conducted during the Central Facilities Committee meeting held on 27 July 2017.

Kindly approve the minutes so that the listed activities may be started at the earliest.

Ashwini K. Agrawal
(Head CRF)

DD(S&P) Chairperson, CFC

Please discuss clause 14. Rest are fine.
3/8/2017

Prof Ashwini Agrawal
HoD(CRF)

Director

As discussed, clause 14 has been revised. Submitted for kind approval.

DD(S&P)/chairperson CFC.
11/8/2017

Recommended for approval
11/8/2017

Director

Approved
14/8/17

Prof. Ashwini Agarwal
Textile Tech

Indian Institute of Technology Delhi

CENTRAL RESEARCH FACILITY

Minutes of Meeting of Central Facility Committee (CFC) held on 27 July 2017

Ref. : **CFC-03-20170731**

Members present: Profs. M Balakrishnan (DDS&P), Ashok Gupta (DDO), B R Mehta (Dean R&D), Sanatanu Roy (Assoc. Dean PG Res), Jayshree Bijwe, Ratnamala Chatterjee, Tapan K Chaudhuri, and Ashwini Agrawal (Head CRF).

The Director also graced the occasion with his presence during the initial part of the meeting.

The following decisions were taken:

1. Head CRF presented the list of equipment being procured (Annexure I) based on the First survey and apprised the members on the status of the equipment purchase. The Director emphasized on completing the equipment purchase process on priority. DD(O) agreed to help in expediting the process at his end.
2. Head CRF also presented the list of equipment desired to be purchased based on 2nd survey. The committee agreed to procure the equipment as presented. The list of approved equipment is enclosed in Annexure II.
3. Committee sanctioned Rs. 65 Cr. for purchase of new equipment as per the 2nd list of equipment. This is in addition of Rs. 20 Cr. already sanctioned for the first list, which is committed for the purchase of equipment under the first list.
4. The committee also approved a budget of Rs. 2.63 Cr under MI01116 for the year 2017-18 for the existing facilities as per the requirements projected by the various facility coordinators (details enclosed in Annexure III). The Cryo probe for 500 MHz NMR was not approved, as its utility across the various departments was not supported by the two CRF surveys. Also the faculty coordinator did not submit any document in support of its usefulness across the departments.
5. The Director and CFC members noted that most of the equipment are for material analysis. It was emphasized that suggestions for high value engineering facilities be obtained from various engineering departments and made part of the CRF facilities.
6. It was also suggested that a few small equipment may be supported at the department level to create department level central facilities on sharing/adoption/co-ownership mode.
7. It was decided to include comprehensive sample preparation facilities for HR TEM.
8. Members agreed to accept the following proposals on co-ownership mode as per CRF policy:
    a. FESEM with EBSD & EDS detectors with joint funding from 1.0 Cr. from AM (department fund), 65 lac from CPSE (project fund) and 1.35 lac from CRF. The space will be provided by AM deptt. The machine time will be

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shared as 4.5 days/week for CRF users, 1 day/week for AM, 0.5 days/week for CPSE. Facility will be jointly managed by two faculty coordinators- One appointed by Head AM and another from PI/nominee from CPSE.

   b. HRTEM with probe corrector with joint funding of Rs. 2 Cr. from Deptt. of Textile Technology (project fund) and will be located in CRF space. The time for CRF users will be at least 5.5 days/week. Faculty coordinator will be PI or his nominee.

9. Members also agreed to accept the proposal of KSBS for adoption of MALDI-TOF/TOF and ESI-MS/MS. The facility is fully funded by KSBS and is located within KSBS. The facility will be made available to CRF users for at least 3 days/week.
10. In order to consolidate functioning of the CRF facilities. It was decided to shift 400 MHz NMR from SMITA lab to CRF space and FESEM from SMITA Lab to CRF electron-microscopy lab behind CCD. On shifting, the available time for CRF users would be increased to 4.2 days/week as per CRF policy.
11. The committee approved the following project positions in addition to the already sanctioned positions under MI01116 for CRF activities
    a. One senior administrative position (to be created) for managing purchases/functioning of CRF
    b. One non technical project position (Project Officer) for managing day-to-day procurement of small/repair parts, management of bills, accounts, etc.
    c. 5 project associate (technical) positions for smooth running of various facilities - Liq. NMR lab, Solid state NMR lab, SEM lab (to relieve Mr. Kuldeep Sharma for software development & management and other administrative activities of CRF), ICP-MS lab, MALDI lab.
    d. One technical project assistant/Jr. project assistant for glass blowing lab
12. The duration of MI01116 was approved to be extended to March 2019 to enable hiring and extension of contracts of the current CRF staff.
13. The committee recommended that a comprehensive requirement for institute-staff positions be projected for CRF. However, most of the other positions for running of the facilities be kept on IRD or institute contracts using highly qualified candidates. A few from these project staff, who spend a few years on contract position (on IRD and institute contract) and show excellent performance can be considered for permanent institute staff positions.
14. The committee also decided to seek funds to financially support a few of the Post Doctoral fellows and PhD scholars registered in different academic departments of the institute. PhD scholars supported through these fellowships will be expected to commit their time towards teaching assistantship to help users in conducting experiments and interpretation of results for specialized facilities such as XPS, SIMS, HRTEM etc.. Similarly, the Post Doctoral candidates will spend part of their time in collaborating with the CRF users and helping them in their experiments do original research. A detailed proposal in this regard will be made and presented to CFC in due course of time.
15. The committee decided that to ensure accountability of institute staff working in CRF labs, their Annual confidential report should be routed through the faculty

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coordinators of the respective CRF labs along with the details of samples/work carried out by them during the year. This process be conveyed to the respective HODs of the concerned staff for implementation.

16. HOD CRF informed the members that as per CRF policy, the tests conducted under CRF are required to be charged. To enable this an accounting software integrated to the booking software has been developed by CRF staff. This software is ready for implementation from the first week of August 2017.
17. Members felt that a CRF contribution of Rs. 2000/semester for research students is very low and needs to be revised upwards. The matter was discussed with the Director and he generously agreed to enhance the CRF contribution as follows:
    a. Rs. 5,000 per student/semester for PhD students for a total of 10 semesters
    b. Rs. 4000 per student/semester for M.Tech./MS(R)/5 year integrated students for a total of 3 semesters
    c. Rs. 3000 per students/semester for B.Tech. Students/M.Sc. for a total of 2 semesters during their final year.
18. The committee also decided to institute annual awards for the best performing CRF facility
19. The committee also decided to institute annual awards to CRF staff showing high performance level.
20. CRF head also apprised the committee that a new building is proposed to be built at Sonipat campus. The committee decided to house all large and sophisticated equipment at Sonipat. The committee also decided to keep small equipment within the campus.
21. Dean R&D informed the members that the institute has contributed Rs. 2 Cr for the establishment of Atom Probe at IIT M and that IIT D users will get 10 h/week for using the facility through remote access. The facility will be listed on CRF website and managed through online booking software of CRF.

The meeting ended with vote of thanks to the Chair.

(M Balakrishanan)	**(Ashwini Kumar Agrawal)**
Chairman, CFC & DD(S&P)	**Head (CRF)**

Director (For approval)

<u>All CFC Members</u>

Annexure I: List of equipment based on 1st survey

- HR TEM (under process, 8 cr + 2 Cr from project)
- FESEM with EBSD & EDS detectors (process nearly complete, 1.35 cr + 0.65 Cr from Project + 1.0 Cr from AM)
- FESEM with STEM and EDS (2.7 Cr)
- XPS (under process, 7 cr)
- High end Thermal Series (TGA, DSC, DMA, TMA, 2 cr)
- FTIR-ATR broad spectra (0.5 cr)
- MALDI, LC/MS (proposed to be adopted from KSBS)

Total cost 22.5 Cr from CRF

Annexure II: List of equipment based on 2nd survey

Name of Instruments	No. of frequent users	Occasional users	Approx.Cost (Rs. cr)
Small angle x-ray scattering (point beam)	18	18	4
Wide angle x-ray diffraction with temp & pressure	21	11	1.5
HRTEM with electron energy loss spectroscopy (EELS)	27	7	20.0
Atomic Force Microscopy (AFM) with modules	25	15	1.2
Scanning Tunnelling Microscopy (STM)	18	19	1.5
Secondary Ion Mass Spectrometer (SIMS-Time of Flight)	18	19	6
Tensile tester with cycling option	17	9	1
Confocal optical microscope	18	19	3
High resolution stylus profilometer	14	13	1
Spectroscopic ellipsometer	7	17	1
3D optical microscope	15	20	.5
Magnetic property measurements- PPM (with VSM and Magnetotransport) or other suitable	9	12	4
Thermal imaging	7	18	1
Impedance spectroscopy	13	11	1
Rotational rheometer	6	15	2
Femto laser cutter	8	15	4
Sample preparation facility and liquid probe for HR TEM	27	7	4

List continued...

Instrument	No. of responses (estimated cost)
Scanning mobility particle sizer/Particle Size Analyzer (1 micron to 1000 microns)	6 (1 cr)
Attachments to Thermogravimetric Analysis- Mass Spectrometry (TGA-MS)/High Pressure TGA (HP-TGA), at least 10 bars pressure or more/In-situ Raman-TGA/DSC/Thermogravimetric Analyser with Gas spectrophotometer and differential thermal analyser	4 (1 cr)
Focused Ion Beam Lithography (FIBL)/ Focused Ion Beam System (Dual Beam)/FIB with 3D EBSD/FIB	4 (4 cr)
high-resolution XRD with wide temperature and pressure range as well as high magnetic field option/High resolution XRD/powder xrd's/X-ray diffractometer for macromolecular crystallography	4
Porosity Measurement System / BET surface area & pore volume analyzer	4 (1 cr)
Analytical Ultracentrifuge/A whole range of refrigerated centrifuges (from table top to large ones)/centrifuge/ultracentrifuges	4 (1 cr)

Total estimated cost of Rs. 65 Cr. (CRF)

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Annexure III Consolidated requirement for existing facilities (* New positions as per minutes)

S. No.	Budget Head (Prices in Rs Lakhs)	TEM (JP)	TEM (JJ)	Cryo-TEM (MB)	SEM (BN)	ICMS (JS)	GB (SD)	NMR (SN)	FESEM (AA)	Raman (AA)	NMR (AA)	LN2 (SjC)	Others*	Total
1	Proposal for any new equipment/ upgradation/accessories (include budgetary quote)	1.8	1	11.5	0	31	0	0	0.8	1.2	0	1		48.3
2	Cost of replacement parts (Only NC parts)	0.07	0	0	0	0	6	0	8	10	4.3	1		29.37
3	AMC or service charges for planned visits	0	0	0	3	0	0	3	4.25	5	6.7	0.5		22.45
4	Consumables	1.3	4	2.63	4	0	27.5	22.7	0.04	1.59	7	7		77.76
5	Manpower (full-time project staff/ part-time student operators)	4.32	4.5	5	10	4	3	4	4.5	4.5	4.5	7	24	79.32
6	Contingency	0.2	1	0	0.5	0	0	0.3	1	0	0	0.5		3.5
7	Workshop/training	0.5	0.5	0	0.5	0	0	1	0	0	0	0		2.5
	Total (Facility-vise)→	8.19	11	19.13	18	35	36.5	31	18.59	22.29	22.5	17	24	263.2

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